Lecture # 2

Theory Of Automata

Lecture#1 Recap

- Definition of the word Automata
- Types of languages, empty/Null String, Alphabets, words, length of strings, Palindromes
- How to form palindromes of even and odd length.

Question

- Q) Prove that there are as many palindromes of length 2n, defined over $\Sigma = \{a,b,c\}$, as there are of length 2n-1, n = 1,2,3… . Determine the number of palindromes of length 2n defined over the same alphabet as well.
- Example:- L={aa,ab,ac,bb,ba,bc,cc,ca,cb} = 9 words.

Solution

- Since the number of symbols are 3 in the alphabet, therefore, the number of length 2n palindromes will be 3^n.

Solution (Cont'd)

Similarly, for palindromes of length 2n-1, we need to calculate their number for each respective symbol in the alphabet, that is, the required number of palindromes are 3^{n-1} for symbol a

Similarly the number of palindromes of length 3n-1, with b or c as middle letter, will be 3^{n-1} as well. Hence the total number of palindromes of length 3n-1 will be :

$$3^{n-1} + 3^{n-1} + 3^{n-1} = 3\,(3^{n-1}) = 3^{n}\,.$$

Kleene Star closure

- In order to further generalize the notation of any combination of strings the famous logician kleene and founder of the 'Theory of Automata' subject has introduced a notation called kleene closure.
- It is denoted by Σ^* and represent all collection of strings defined over Σ including Null string.

- The language produced by Kleene closure is infinite. It contains infinite words, however each word has finite length.

Examples

- If $\Sigma = \{x\}$

  Then $\Sigma^* = \{\Lambda, x, xx, xxx, xxxx, ...\}$

- If $\Sigma = \{0,1\}$

  Then $\Sigma^* = \{\Lambda, 0, 1, 00, 01, 10, 11, ...\}$

- If $\Sigma = \{aaB, c\}$

  Then $\Sigma^* = \{\Lambda, aaB, c, aaBaaB, aaBc, caaB,$
  $cc, ...\}$

Q) Consider the language S*, where S = {a b}. How many words does this language have of length 2? of length 3? of length n?

A) Number of words = n^m

Length 2: $2^2 = 4$

Length 3: $2^3 = 8$

Length n: 2^n

- Consider the language S*, where S = {aa b}. How many words does this language have of length 4? of length 5? of length 6? What can be said in general?

Words of length 0: Λ = 1 word

Words of length 1: b = 1 word

Words of length 2: (Add aa to all words of length 0 --> 0 + 2 = 2)

Add b to all words of length 1 --> 1 + 1 = 2)

aa bb = 2 words

Words of length 3:
(Add aa to all words of length 1 --> 1 + 2 = 3)
(Add b to all words of length 2 --> 2 + 1 = 3)
aab baa bbb = 3 words

Words of length 4:
(Add b to all words of length 3)
(Add aa to all words of length 2)
baab bbaa bbbb aaaa aabb = 5 words

Words of length 5:
(Add aa to the 3 words of length 3)
(Add b to the 5 words of length 4) = 8 words

Words of length 6:

(Add aa to the 5 words of length 4)

(Add b to the 8 words of length 5) = 13 words

This is the Fibonacci sequence: 1,2,3,5,8,13,…

Plus Operation

- With Plus Operation, combination of different letters are formed. However, Null String is not part of the generated language.

Examples

- If $\Sigma = \{x\}$

  Then $\Sigma^+ = \{ x, xx, xxx, xxxx, \ldots.\}$
- If $\Sigma = \{0,1\}$

  Then $\Sigma^+ = \{ 0, 1, 00, 01, 10, 11, \ldots.\}$
- If $\Sigma = \{aaB, c\}$

  Then $\Sigma^+ = \{aaB, c, aaBaaB, aaBc, caaB, cc, \ldots.\}$

Lecture#2 Summary

- Kleene Closure
- Examples of Kleene Closure.
- What is Plus Operation
- Plus Operation examples